

PROJECT 1

Group Project Task List

Student Name	Student ID	Role
Qwyn Celine Djimondo	001202400205	Team Leader
Tentin Nofa Tuzika	001202400134	Member 1
Tessalonika Angeline Purba	001202400210	Member 2

Chapter 1. Introduction

As a group, identify potential problems in your daily life that can be solved using programming.

“Menstrual Cycle Tracker and Reminder”

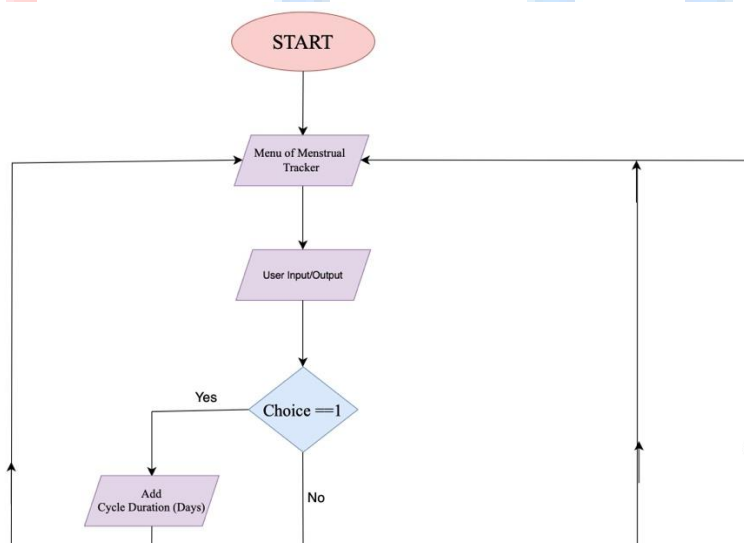
Many women find it difficult to track their menstrual cycles, especially irregular ones. This Menstrual Cycle Tracker and Reminder program aims to help users record how many days their period lasts, predict the next cycle, send reminders before the cycle starts, record symptoms or moods, and read health information related to menstruation.

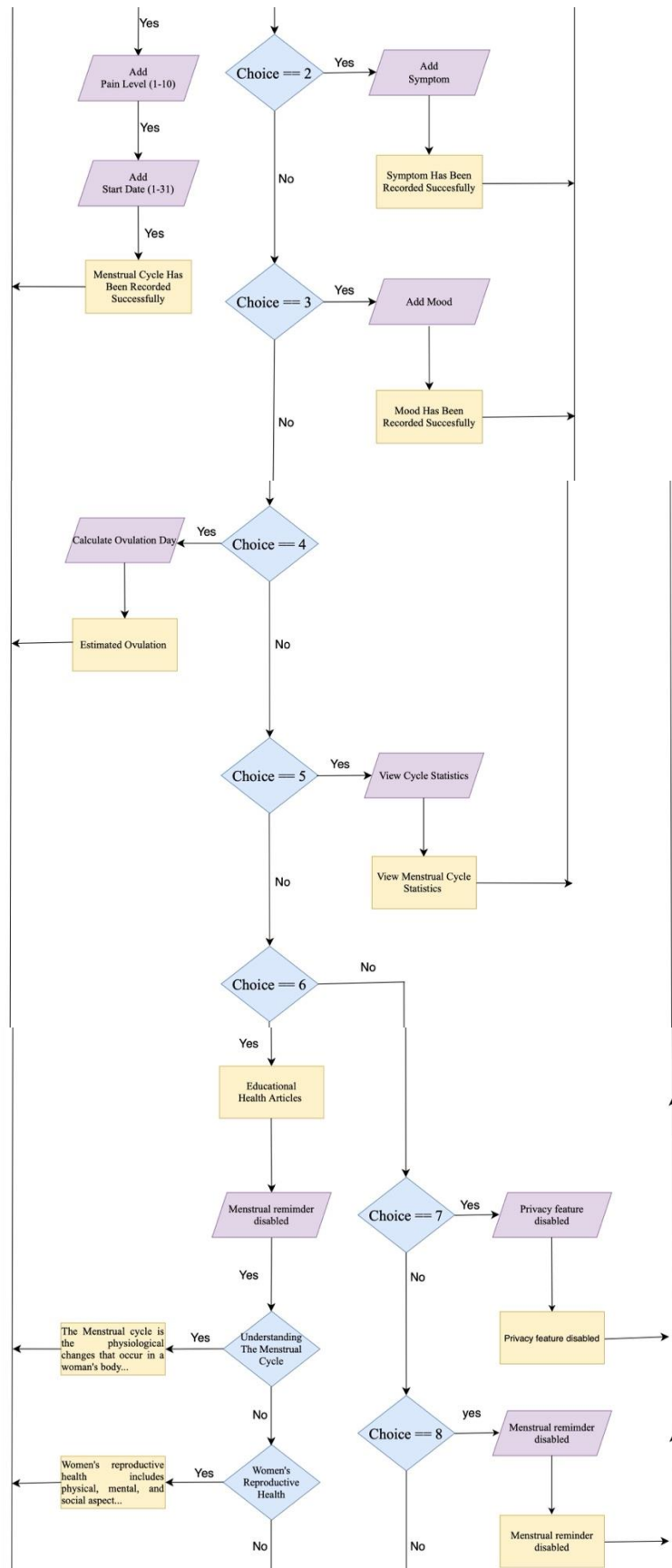
Features:

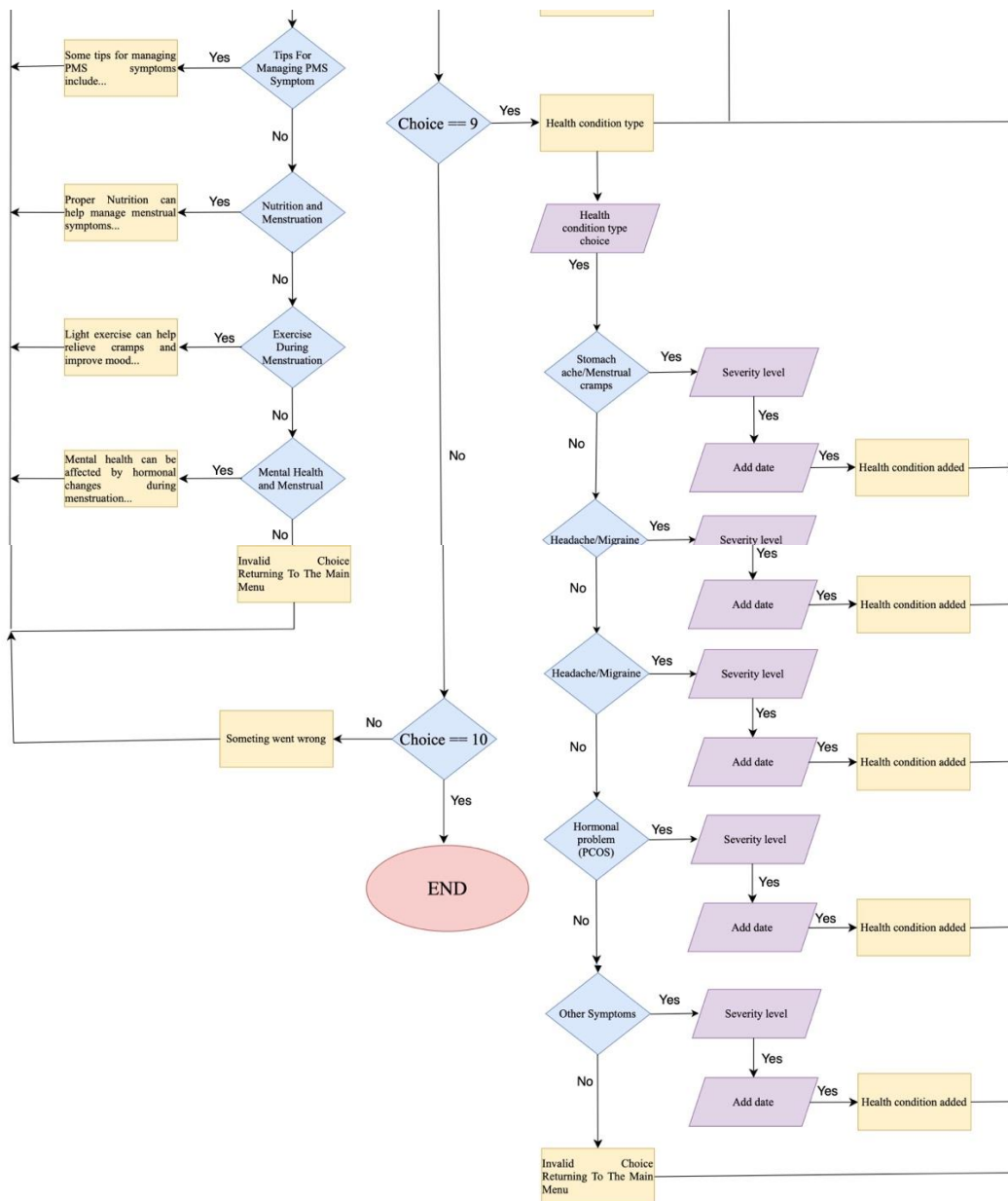
1. Enable users to record menstrual cycles
2. Enable users to add symptoms
3. Enable users to add moods
4. Enable users to calculate ovulation days
5. Enable users to view cycle statistics
6. Enable users to read health education articles
7. Enable users to enable/disable data privacy
8. Enable users to enable/disable menstrual reminders
9. Enable users to add for health conditions

Chapter 2. Flowchart

on the description you provided in Chapter 1, provide the end-to-end flowchart of your program







Chapter 3: The Program

Provide the program in this section (copy and paste your program here)

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Line	Code: “Menstrual Cycle Tracker and Reminder”
1	#include <iostream>
2	#include <string>
3	#include <iomanip>
4	#include <ctime>
5	
6	using namespace std;
7	
8	class MenstrualTracker {
9	private:
10	static const int MAX_RECORDS = 100; // Maximum number of records
11	
12	int cycleDetails[MAX_RECORDS][3]; // Stores cycle details: [duration, pain level,
13	start date] (2D array)
14	string symptoms[MAX_RECORDS]; // Stores symptoms (1D array)
15	string moods[MAX_RECORDS]; // Stores mood states (1D array)
16	int healthHistory[MAX_RECORDS][2][3]; // Tracks specific health data (3D array:
17	type, severity, date)
18	
19	int cycleCount; // Tracks the number of recorded cycles
20	int symptomCount; // Tracks the number of recorded symptoms
21	int moodCount; // Tracks the number of recorded moods
22	int healthHistoryCount; // Tracks the number of recorded health histories
23	
24	int lastCycleDuration; // Duration of the last cycle
25	bool dataPrivacyEnabled; // Simulates privacy feature
26	bool reminderEnabled; // Simulates <u>menstrual</u> reminders
27	
28	// Function to get health type description
29	string getHealthTypeDescription(int type) {
30	switch (type) {
31	case 1: return "Stomach ache/ <u>menstrual</u> cramps";
32	case 2: return "Headache/ <u>migraine</u> ";
33	case 3: return " <u>Hormonal</u> problems (PCOS);
34	case 4: return "Sleep disorders";
35	case 5: return "Other symptoms";
36	default: return "Health type unknown";
37	}
38	}
39	
40	public:
41	MenstrualTracker()
42	: cycleCount(0), symptomCount(0), moodCount(0), healthHistoryCount(0),
43	lastCycleDuration(28), dataPrivacyEnabled(true), reminderEnabled(true) {}
44	
45	// Function to record <u>menstrual</u> cycle
46	void trackCycle(int days, int painLevel, int startDate) {
47	if (cycleCount < MAX_RECORDS) {
48	cycleDetails[cycleCount][0] = days; // Cycle duration
49	cycleDetails[cycleCount][1] = painLevel; // Pain level
50	cycleDetails[cycleCount][2] = startDate; // Start date
51	cycleCount++;
52	lastCycleDuration = days;

```

53     cout << "Cycle with duration " << days
54         << " days, pain level " << painLevel
55         << ", and start date " << startDate
56         << " has been recorded successfully.\n";
57     } else {
58         cout << "Cycle records are full. Unable to record more.\n";
59     }
60 }
61
62 // Function to add symptoms experienced by the user
63 void addSymptom(const string& symptom) {
64     if (symptomCount < MAX_RECORDS) {
65         symptoms[symptomCount++] = symptom;
66         cout << "Symptom \"" << symptom << "\" has been recorded
67 successfully.\n";
68     } else {
69         cout << "Symptom records are full. Unable to record more.\n";
70     }
71 }
72
73 // Function to add the user's mood
74 void addMood(const string& mood) {
75     if (moodCount < MAX_RECORDS) {
76         moods[moodCount++] = mood;
77         cout << "Mood \"" << mood << "\" has been recorded successfully.\n";
78     } else {
79         cout << "Mood records are full. Unable to record more.\n";
80     }
81 }
82
83 // Function to add health condition
84 void addHealthHistory(int type, int severity, int date) {
85     if (healthHistoryCount < MAX_RECORDS) {
86         healthHistory[healthHistoryCount][0][0] = type; // Health type
87         healthHistory[healthHistoryCount][0][1] = severity; // Severity level
88         healthHistory[healthHistoryCount][0][2] = date; // Date recorded
89         healthHistoryCount++;
90         cout << "Health condition added: Type " << type
91             << " (" << getHealthTypeDescription(type) << "), "
92             << "Severity " << severity << ", Date " << date << ".\n";
93     } else {
94         cout << "Health condition records are full. Unable to record more.\n";
95     }
96 }
97
98 // Function to calculate ovulation day
99 void calculateOvulation() {
100     if (cycleCount == 0) {
101         cout << "No cycle data recorded yet.\n";
102         return;
103     }
104
105     int totalCycleDays = 0;
106     for (int i = 0; i < cycleCount; i++) {
107         totalCycleDays += cycleDetails[i][0];
108     }

```

```

109
110     int averageCycle = totalCycleDays / cycleCount;
111     int ovulationDay = averageCycle - 14;
112     cout << "Estimated ovulation day: Day " << ovulationDay << " of the cycle.\n";
113 }
114
115 // Function to display menstrual cycle statistics
116 void displayStatistics() {
117     if (cycleCount == 0) {
118         cout << "No statistics data to display yet.\n";
119         return;
120     }
121
122     int totalDays = 0;
123     int maxCycle = cycleDetails[0][0];
124     int minCycle = cycleDetails[0][0];
125
126     for (int i = 0; i < cycleCount; i++) {
127         totalDays += cycleDetails[i][0];
128         if (cycleDetails[i][0] > maxCycle) maxCycle = cycleDetails[i][0];
129         if (cycleDetails[i][0] < minCycle) minCycle = cycleDetails[i][0];
130     }
131
132     double averageCycle = static_cast<double>(totalDays) / cycleCount;
133
134     cout << "\n--- Menstrual Cycle Statistics ---\n";
135     cout << "Number of recorded cycles: " << cycleCount << "\n";
136     cout << "Average cycle duration: " << fixed << setprecision(2) << averageCycle
137 << " days.\n";
138     cout << "Longest cycle duration: " << maxCycle << " days.\n";
139     cout << "Shortest cycle duration: " << minCycle << " days.\n";
140 }
141
142 // Function to display educational health articles
143 void displayEducationalArticles() {
144     cout << "\n--- Educational Health Articles ---\n";
145     cout << "1. Understanding the Menstrual Cycle\n";
146     cout << "2. Women's Reproductive Health\n";
147     cout << "3. Tips for Managing PMS Symptoms\n";
148     cout << "4. Nutrition and Menstruation\n";
149     cout << "5. Exercise During Menstruation\n";
150     cout << "6. Mental Health and Menstruation\n";
151     cout << "Enter article number to read more (0 to go back): ";
152
153     int articleChoice;
154     cin >> articleChoice;
155
156     switch (articleChoice) {
157         case 1:
158             cout << "\n--- Understanding the Menstrual Cycle ---\n";
159             cout << "The menstrual cycle is the physiological changes that occur in a
160 woman's body...\n";
161             break;
162         case 2:
163             cout << "\n--- Women's Reproductive Health ---\n";
164             cout << "Women's reproductive health includes physical, mental, and

```

```

165     social aspects...\n";
166         break;
167     case 3:
168         cout << "\n--- Tips for Managing PMS Symptoms ---\n";
169         cout << "Some tips for managing PMS symptoms include...\n";
170         break;
171     case 4:
172         cout << "\n--- Nutrition and Menstruation ---\n";
173         cout << "Proper nutrition can help manage menstrual symptoms...\n";
174         break;
175     case 5:
176         cout << "\n--- Exercise During Menstruation ---\n";
177         cout << "Light exercise can help relieve cramps and improve mood...\n";
178         break;
179     case 6:
180         cout << "\n--- Mental Health and Menstruation ---\n";
181         cout << "Mental health can be affected by hormonal changes during
182 menstruation...\n";
183         break;
184     case 0:
185         cout << "Returning to the main menu...\n";
186         break;
187     default:
188         cout << "Invalid choice. Returning to the main menu.\n";
189         break;
190     }
191 }
192
193 // Function to toggle data privacy
194 void toggleDataPrivacy() {
195     dataPrivacyEnabled = !dataPrivacyEnabled;
196     cout << "Data privacy has been " << (dataPrivacyEnabled ? "enabled" :
197 "disabled") << ".\n";
198 }
199
200 // Function to toggle reminder
201 void toggleReminder() {
202     reminderEnabled = !reminderEnabled;
203     cout << "Menstrual reminder has been " << (reminderEnabled ? "enabled" :
204 "disabled") << ".\n";
205 }
206
207 // Simple user interface
208 void userFriendlyInterface() {
209     cout << "\n--- Menstrual Tracker ---\n";
210     cout << "1. Record menstrual cycle\n";
211     cout << "2. Add symptom\n";
212     cout << "3. Add mood\n";
213     cout << "4. Calculate ovulation day\n";
214     cout << "5. View cycle statistics\n";
215     cout << "6. Read educational health articles\n";
216     cout << "7. Enable/disable data privacy\n";
217     cout << "8. Enable/disable menstrual reminder\n";
218     cout << "9. Add health condition\n";
219     cout << "10. Exit\n";
220 }

```

```

221     };
222
223     int main() {
224         MenstrualTracker tracker;
225         int choice, days, painLevel, startDate, type, severity, date;
226         string symptom, mood;
227
228         do {
229             tracker.userFriendlyInterface();
230             cout << "Choose an option: ";
231             cin >> choice;
232
233             switch (choice) {
234                 case 1:
235                     cout << "Enter cycle duration (days): ";
236                     cin >> days;
237                     cout << "Enter pain level (1-10): ";
238                     cin >> painLevel;
239                     cout << "Enter start date (1-31): ";
240                     cin >> startDate;
241                     tracker.trackCycle(days, painLevel, startDate);
242                     break;
243                 case 2:
244                     cout << "Enter symptom: ";
245                     cin.ignore();
246                     getline(cin, symptom);
247                     tracker.addSymptom(symptom);
248                     break;
249                 case 3:
250                     cout << "Enter mood: ";
251                     cin.ignore();
252                     getline(cin, mood);
253                     tracker.addMood(mood);
254                     break;
255                 case 4:
256                     tracker.calculateOvulation();
257                     break;
258                 case 5:
259                     tracker.displayStatistics();
260                     break;
261                 case 6:
262                     tracker.displayEducationalArticles();
263                     break;
264                 case 7:
265                     tracker.toggleDataPrivacy();
266                     break;
267                 case 8:
268                     tracker.toggleReminder();
269                     break;
270                 case 9:
271                     cout << "Enter health type (choose from below):\n";
272                     cout << "1. Stomach ache/menstrual cramps\n";
273                     cout << "2. Headache/migraine\n";
274                     cout << "3. Hormonal problems (PCOS)\n";
275                     cout << "4. Sleep disorders\n";
276                     cout << "5. Other symptoms\n";
277

```


278	cout << "Your choice: ";
279	cin >> type;
280	cout << "Enter severity level (1-10): ";
281	cin >> severity;
282	cout << "Enter date (1-31): ";
283	cin >> date;
284	tracker.addHealthHistory(type, severity, date);
285	break;
286	case 10:
287	cout << "Exiting the program.\n";
288	break;
289	default:
290	cout << "Invalid choice. Please try again.\n";
291	}
292	} while (choice != 10);
293	
294	return 0;
295	}
296	
297	
298	
299	
300	
301	
302	
303	
304	
305	
306	
307	
308	
309	



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Chapter 4: The Screenshot

Give a screenshot of your feature here. Each screenshot should represent 1 feature, as listed in Chapter 1.

1st Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 1
Enter cycle duration (days): 7
Enter pain level (1-10): 8
Enter start date (1-31): 2
Cycle with duration 7 days, pain level 8, and start date 2 has been recorded successfully.
```

2nd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 2
Enter symptom: stomach ache
Symptom "stomach ache" has been recorded successfully.
```

3rd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 3
Enter mood: sad and angry
Mood "sad and angry" has been recorded successfully.
```

4th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 4
Estimated ovulation day: Day -7 of the cycle.
```

5th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 5

--- Menstrual Cycle Statistics ---
Number of recorded cycles: 1
Average cycle duration: 7.00 days.
Longest cycle duration: 7 days.
Shortest cycle duration: 7 days.
```

6th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 6

--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation
Enter article number to read more (0 to go back): 1

--- Understanding the Menstrual Cycle ---
The menstrual cycle is the physiological changes that occur in a woman's body...
```

7th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 2

--- Women's Reproductive Health ---

Women's reproductive health includes physical, mental, and social aspects...

8th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 3

--- Tips for Managing PMS Symptoms ---

Some tips for managing PMS symptoms include...

```

--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

```

Choose an option: 6

```

--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

```

Enter article number to read more (0 to go back): 4

```

--- Nutrition and Menstruation ---

```

Proper nutrition can help manage menstrual symptoms...

```

--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

```

Choose an option: 6

```

--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

```

Enter article number to read more (0 to go back): 5

```

--- Exercise During Menstruation ---

```

Light exercise can help relieve cramps and improve mood...

11th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 6

--- Mental Health and Menstruation ---

Mental health can be affected by hormonal changes during menstruation...

12th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 7

Data privacy has been disabled.

13th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 8

Menstrual reminder has been disabled.

14th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 9
Enter health type (choose from below):
1. Stomach ache/menstrual cramps
2. Headache/migraine
3. Hormonal problems (PCOS)
4. Sleep disorders
5. Other symptoms
Your choice: 1
Enter severity level (1-10): 9
Enter date (1-31): 2
Health condition added: Type 1 (Stomach ache/menstrual cramps), Severity 9, Date 2.
```

15th Feature Cods

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit
Choose an option: 9
Enter health type (choose from below):
1. Stomach ache/menstrual cramps
2. Headache/migraine
3. Hormonal problems (PCOS)
4. Sleep disorders
5. Other symptoms
Your choice: 2
Enter severity level (1-10): 6
Enter date (1-31): 2
Health condition added: Type 2 (Headache/migraine), Severity 6, Date 2.
```

16th Feature Cods

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 9

Enter health type (choose from below):

1. Stomach ache/menstrual cramps
2. Headache/migraine
3. Hormonal problems (PCOS)
4. Sleep disorders
5. Other symptoms

Your choice: 3

Enter severity level (1-10): 4

Enter date (1-31): 2

Health condition added: Type 3 (Hormonal problems (PCOS)), Severity 4, Date 2.

17th Feature Cods

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 9

Enter health type (choose from below):

1. Stomach ache/menstrual cramps
2. Headache/migraine
3. Hormonal problems (PCOS)
4. Sleep disorders
5. Other symptoms

Your choice: 4

Enter severity level (1-10): 7

Enter date (1-31): 2

Health condition added: Type 4 (Sleep disorders), Severity 7, Date 2.

18th Feature Cods

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 9

Enter health type (choose from below):

1. Stomach ache/menstrual cramps
2. Headache/migraine
3. Hormonal problems (PCOS)
4. Sleep disorders
5. Other symptoms

Your choice: 5

Enter severity level (1-10): 5

Enter date (1-31): 2

Health condition added: Type 5 (Other symptoms), Severity 5, Date 2.

19th Feature Cods

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminder
9. Add health condition
10. Exit

Choose an option: 10

Exiting the program.

Chapter 5: Requirement checks

No	Requirement	Yes/No, Line No X - Y
1	Have at least one selection.	Yes Switch Line 234-292
2	Have at least one loop.	Yes Do-While Line 229-293
3	Have at least one 1 Dimensional Array.	Yes Line 14 & 15
4	Have at least one 2 Dimensional Array	Yes Line 12
5	Have at least one 3 Dimensional Array	Yes Line 16
6	Have at least one mathematical operator calculation	Yes Line 110-111
7	Have at least one Boolean expression	Yes Line 195 & 202

Chapter 6: Log Hours

<i>Student Name</i>	<i>Activity</i>	<i>Log Hours</i>
Qwyn Celine Djimondo	- Discussion/Planning - Report/Documentation - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 20 minutes - 30 minutes - 20 minutes - 45 minutes
Tentin Nofa Tuzika	- Discussion/Planning - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 20 minutes - 20 minutes - 1 hour
Tessalonika Angeline Purba	- Discussion/Planning - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 20 minutes - 20 minutes -30 minutes